CLASS IX • SCIENCE

Max Marks: 40
Time Allowed: 90 Mins
Pattern: NCERT & Exemplar

Chapter 6: Tissues (Advanced HOTS Level)

HIGH ORDER THINKING SKILLS (HOTS) • CONCEPTUAL TEST PAPER

General Instructions:

- All questions are compulsory. Focus on scientific reasoning and analytical explanations.
- Section A: Assertion-Reasoning & Analytical VSA (1 Mark Each).
- Section B: Application-Based Short Answers (2 Marks Each).
- Section C: Comparative & Structural Analysis (3 Marks Each).
- Section D: Comprehensive & Experimental Long Answers (5 Marks Each).
- Section E: Case-Based Scenario Analysis (4 Marks).

SECTION A — Analytical & Assertion-Reasoning (1 Mark Each)

1. A plant cell possesses a dense cytoplasm, a prominent nucleus, but lacks vacuoles. Identify the specific tissue type and give a reason for the absence of vacuoles. **[1]**

2. Why does a desert plant like a cactus have a significantly thicker epidermis coated with cutin compared to a aquatic plant? **[1]**

3. **Assertion (A):** Cardiac muscles do not get fatigued throughout a person's lifespan.
Reason (R): Cardiac muscle cells are branched, cylindrical, uninucleate, and contain a rich blood supply for continuous aerobic respiration.
(a) Both A and R are true and R is correct explanation of A. (b) Both A and R are true but R is not correct explanation. (c) A is true, R is false. (d) A is false, R is true. **[1]**

4. If the tip of a growing onion root is clipped off, root growth halts completely. Which exact meristematic tissue was removed? **[1]**

5. A matrix contains fluid blood plasma with dissolved proteins, salts, and hormones. Why is blood classified as a "connective" tissue despite lacking structural fibers in its matrix? **[1]**

SECTION B — Conceptual & Application Questions (2 Marks Each)

6. A farmer notices that young stems and leaves of a plant bend easily during high winds without breaking. Name the mechanical tissue responsible and explain its structural adaptation. **[2]**

7. What would happen to a plant if the phloem tissue at the base of a stem is girdled (gently removed in a ring) while xylem remains completely intact? Explain the immediate and long-term consequences. **[2]**

8. Identify the type of epithelial tissue that line: **[2]**

(a) The respiratory tract where mucus needs to be pushed forward continuously.

(b) Kidney tubules where reabsorption occurs and mechanical support is required.

9. How does chemical deposition of **Suberin** in cork cells and **Lignin** in sclerenchyma cells affect their permeability to water and gases? **[2]**

SECTION C — Scientific Justification & Comparison (3 Marks Each)

10. An athlete suffers an injury while playing football. The doctor confirms a tear in a connective tissue that connects his thigh muscle to the knee bone, but no bone breakage. **[3]**

(a) Name the tissue involved in this tear.

(b) Contrast this tissue with the tissue that connects bone to bone in terms of matrix flexibility and fiber density.

11. Explain how complex permanent tissues (Xylem and Phloem) differ from simple permanent tissues (Parenchyma, Collenchyma, Sclerenchyma) in terms of cellular organization and coordination. **[3]**

12. Compare smooth (unstriated) muscles and skeletal (striated) muscles based on: **[3]**

(a) Shape and nuclear arrangement of individual cells.

(b) Voluntary vs. involuntary control mechanisms.

(c) Presence or absence of alternating light and dark bands.

SECTION D — Comprehensive Analytical Questions (5 Marks Each)

13. Regarding the nervous system and signal transmission: **[5]**

(a) Draw a structural diagram of a single neuron labeling Cyton, Axon, Dendrite, and Nerve Ending. **(2 marks)**

(b) Trace the directional flow of an electrical impulse within a single neuron from reception to transmission. **(2 marks)**

(c) Name the tissue that protects and insulates nerve fibers. **(1 mark)**

14. As a tree grows older, its outer stem layer undergoes significant structural transformation to form cork/bark: **[5]**

(a) Explain the origin of cork cambium (secondary meristem) in woody plants. **(1.5 marks)**

(b) Describe three key structural characteristics of mature cork cells. **(1.5 marks)**

(c) How does suberization in cork cells benefit the plant against physical and biological stress? **(2 marks)**

SECTION E — Case-Based High Order Thinking Scenario (4 Marks)

15. Read the scenario carefully and answer the follow-up questions: **[4]**

> During a histology laboratory practical, a student observes three unidentified animal tissue slides under a microscope:
> • **Slide A:** Single layer of column-like tall cells with cilia on their free surface.
> • **Slide B:** Cells embedded in a hard, non-flexible matrix made of Calcium and Phosphorus compounds arranged in concentric rings.
> • **Slide C:** Fat-storing cells filled with lipid globules located beneath the skin and around internal organs.

(a) Identify the exact tissue present in **Slide B** and state its primary structural function in the body. **(1 mark)**

(b) Identify the tissue in **Slide C**. How does it act as an insulator for warm-blooded animals? **(1 mark)**

(c) Predict the organ location for **Slide A** and explain why the presence of cilia is essential for that specific location. **(2 marks)**